

Machine Learning (EE526)

Mid Semester Exam

04/03/2023

Full Marks: 100

Time: 2 Hours

[Q1] Compute the Weighted Median from the following weighted dataset (X, W) [10]

X	4	-6	0	8	-2	10	-4	2	-8	6
W	4	2	3	2	2	4	3	4	3	3

[Q2] Compute the covariance matrix from the following five 2-D vectors. [10]

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ 9 \end{bmatrix}, \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

[Q3] Compute the Mahalanobis Distance between the following point-sets S_1 and S_2 . [10]

S₁	-4	4	-2	0	2
S₂	8	10	9	14	13

[Q4] Perform linear regression to fit the line $Y = a + bX$ on the following set of points (X, Y) and estimate the values of a and b . [20]

X	-1.313	8.585	5.353	10.00	-8.383
Y	-1.062	39.506	27.089	46.69	-28.019

[Q5] Consider a dataset containing 17.7% outliers. We want to learn a model from this dataset that requires 10 data instances. What is the expected number of RANSAC iterations that we should perform? Deduce necessary expressions. [3+7=10]

[Q6] Suggest a Threshold T for the dataset $S = \{2, 30, 34, 5, 62, 6, 64, 37, 3, 15\}$ using Otsu's method. Start iterations with $T_0 = 6$. [10]

- ✓[Q7] Perform K-Means clustering for the dataset **X** and Fill up the following table while estimating Labels **L** and Centroids **M** for each iteration. Iterate till convergence for the specified number of maximum iterations. Report the final labels and the cluster centroids in the following tabular form.

[20]

#	X										M₁	M₂	M₃
	-3	25	29	0	57	1	59	32	-2	10			
L₀	1	1	1	2	2	2	3	3	3	1			
L₁													

- ✓[Q8] The following function **f(v)** is to be minimized.

$$f(v) = \frac{(v - 110)^2}{110}$$

Suggest the objective function **y = g(x)** for solving this problem using stochastic search. We are solving this problem using a population size of **n = 5**. Consider the solution population **S_{t-1}** for **g(x)** as **S_{t-1} = {76, 82, 132, 155, 124}**. Find the number of children that will be generated from each of these solutions using the process of random walk.

[10]